

Pyrolysis of Polyolefin-Enriched Mixed Plastic Waste Streams: Effects of Pretreatments and Presence of Hydrogen during Pyrolysis

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ABSTRACT: Pyrolysis is considered a promising method for the chemical recycling of mixed plastic waste streams, yet systematic experimental evaluations of these mixed streams remain limited. Herein, the pyrolysis of a polyolefin-enriched (PO-enriched) waste stream derived from a DKR-350 mixed plastic waste is investigated. Particularly, the influence of an additional dry-washing step and the presence of hydrogen during pyrolysis on product yield and quality is studied. Dry-washing to reduce surface contamination in mixed plastic waste is an application of a loop dryer to improve the quality of mixed plastic waste. Batch pyrolysis in an autoclave reactor at 420 °C for 1 h under nitrogen resulted in up to 78% oil yield (on material intake) for dry-washed DKR-350. Product oil composition was characterized by using elemental analysis and GC × GC-TOFMS, revealing a high aliphatic hydrocarbon content (up to 94%). Moreover, pyrolysis oils from PO-enriched DKR-350 contained significant amounts of linear and branched aliphatic components (up to 80%). Dry-washing reduced inorganic feedstock contamination (e.g., pigments, salts) to below 4%, enhancing oil yield by 10% as well as increasing the overall hydrogen and carbon to oil yields. Remarkably, unwashed DKR-350-derived pyrolysis oil had low chlorine levels (as low as 17 ppm) compared to 32 ppm for oils from dry-washed streams. Experiments in the presence of hydrogen instead of nitrogen resulted in product oils with a higher H/C ratio, indicating hydrogen incorporation in the product during pyrolysis. For example, the level of unsaturated hydrocarbons reduced from 25% to 20%, and the heteroatom content was reduced by up to 31%. Overall, high liquid and carbon yields are obtained for pyrolysis oils from contaminated plastic waste, which to the best of our knowledge have not been reported before. The pretreatments and processing methods outlined in this work positively affect product properties, reducing the need for post-treatment processing.

Dual approach for improving yield (1) and composition (2) of pyrolysis oil

Feedstock	Pretreatment	Pyrolysis (batch)	Oil
(ref)	None	N ₂	71%
	(1) Dry-washing	N ₂	+ 10% oil yield
	(2) None	H ₂	- 30% oxygen in oil

1. INTRODUCTION

With ever-increasing plastic consumption in society and the pressing need to increase renewable and recycled content, the development of new and improved plastic recycling methods is paramount. Chemical recycling via pyrolysis to produce oils that can serve as carbon feedstocks for synthesizing new plastics through petrochemical routes has emerged as a particularly promising option for polyolefins (PO) such as polyethylene (PE), polypropylene (PP), and polystyrene (PS).^{1,2} However, a significant challenge in the pyrolysis of mixed plastic wastes rich in PO is their inhomogeneous nature and the presence of a variety of other polymers, such as polyethylene terephthalate (PET) and polyvinyl chloride (PVC),^{3,4} which can negatively impact liquid yields and increase impurity levels in the products.⁵ Mechanical recycling remains an option for clean, single-material plastic products whereas condensation polymers (PET and PA) are chemically recyclable through hydrolysis, ammonolysis, or alcoholysis.^{6,7} However, mixed plastic waste streams rich in polyolefins are generally unsuitable for these methods, making them prime candidates for chemical recycling via pyrolysis.¹ These mixed plastics often contain composites, such as multimaterial or

multilayer films, making them difficult to separate.⁸ Additionally, mixed plastics are frequently contaminated with moisture, food residues, biomass, salts, and other substances, resulting in the presence of various metals and halogens.⁹ Removing these contaminants before pyrolysis is crucial as elements like sulfur, chlorine, and certain metals can corrode equipment and poison catalysts during downstream processing in refineries.¹⁰

Life cycle and techno-economic assessments for mixed waste plastic pyrolysis often consider only clean streams devoid of metals, paper, or biomass residues, which does not reflect the actual nature of mixed plastic waste.^{11–13} Experimental studies frequently use rigorously sorted, cleaned, and pretreated plastic waste for pyrolysis, often still labeled as “mixed plastic waste”.^{14–17} Alternatively, very clean packaging waste streams

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Table 1. Overview of Mixed Plastic (Wastes), Their Elemental Composition, Pyrolysis Oil Yield, and Elemental Composition of the Oil Product

plastic feedstock composition	elemental composition				oil yield (%)	elemental composition of oil				ref
	C (%)	H (%)	N (%)	O (%)		C (%)	H (%)	N (%)	O (%)	
DKR-350 mixed plastic waste	73.3	10.1	0.6	16.1 ^a	65.9	75.2	10.0	0.3	14.5 ^a	[35]
DKR-350 mixed plastic waste (Hot water washed)	76.0	11.4	0.4	12.3 ^a	69.1	80.6	11.1	0.2	8.2 ^a	
recycled PP, PE and PS						85.0	13.3	0.3	1.2	[36]
washed and dried mixed plastics with PET and PVC manually removed (73% PE, 10.4% PP, 16.6% PS)	86.7 ^b	13.3 ^b	0 ^b	0 ^b	29 (oil) 50.1 (tar)	88.8 89.2	7.2 16.2	0.2 0.04	3.8 ^a 0 ^a	[17]
virgin plastic mixture (70% PP, 10% PE, 5% EVOH, 5% PET, 10% PS), best result of duplicate	84.9 ^{b,c}	13.2 ^{b,c}	0 ^b	1.9 ^{b,c}	~83	~86	~8	~0.5	~5.5 ^{a,b}	[30]
MPO rigids (46% PP, 53% PE, 1% others)	85.3	14.2	0.2	0.3	89	86.2	13.7	0.1	<0.1	[34]
postconsumer waste PE films (97% PE, 1% PP, ~2% others)	84.3	13.9	0.2	1.7	85	85.3	14.5	0.004	0.2	
PO-rich dry-washed DKR-350, H ₂	76.4–79.8 ^d	12.4–12.9 ^d	~1 ^d	6.7–8.2 ^d	76.6	79.0	12.9	0.6	5.7	this work
PO-rich unwashed DKR-350, H ₂					68.5	81.5	13.1	0.5	4.4	

^aBy difference. ^bEstimate from composition. ^cAssuming 12% vinyl alcohol (VOH) in EVOH. ^dBy back-calculation of products.

containing high polyolefin content are used as end-of-life mixed plastic waste,^{18,19} but these do not represent realistic postconsumer mixed plastic waste. Bridging the gap between realistic mixed plastic waste and suitable pyrolysis feedstock requires extensive pretreatment that can operate at large scales to be able recycling with virgin plastics.²⁰

Extensive research has demonstrated the feasibility of plastic pyrolysis using clean and virgin plastics, with well-understood effects of process conditions such as temperature, reaction time, and catalyst addition on product yields and molecular composition.^{21–23} Pyrolysis of polyolefins yields highest amounts of liquid at slow heating rates and moderate temperatures (below 500 °C),²⁴ with batch operated stirred tank reactors being particularly suitable for this method.^{23,25} Liquid yields from virgin PO feeds can reach up to ~95%,^{26,27} with optimal temperatures reported between 350 and 500 °C.² Slow pyrolysis typically spans several hours, whereas fast pyrolysis methods, such as fluidized beds, tend to produce more gaseous products and secondary products like aromatics at higher temperatures.^{24,28,29} Waste plastics typically have lower oil yields than pure polyolefins due to the presence of nonplastic materials like biomass or condensation polymers like PET.^{30–32}

Pyrolysis of mixed plastic waste aims to produce an oil product for a steam cracker to produce new, virgin quality building blocks.¹ However, the impurities in the feedstock propagate to the pyrolysis oil and these impurities exceed industrial thresholds for petrochemical processes.^{33,34} Table 1 shows the plastic composition, elemental composition of the feedstock, oil yield, and the elemental composition of several pyrolysis feedstocks and oils. Even with washing and sorting, high heteroatom contents are still expected in pyrolysis oil and oil yields for contaminated plastic waste are lower than those for clean plastics. Previous work by Genuino et al. already showed that DKR-350 mixed plastic waste contains up to 16.1% oxygen, which is reduced to 12.3% after washing with hot water.³⁵ The pyrolysis oils they obtained subsequently also contained large amounts of oxygen (14.5% and 8.2%, respectively). Other researchers have similarly found that even with near-perfect sorting, completely removing oxygen from the pyrolysis oil is unattainable,^{17,36} especially if industrial thresholds are around 100 ppm for oxygen and nitrogen.³³

Oxygen and nitrogen containing compounds are traced back to, for example, EVOH (ethyl vinyl alcohol) or polyamides present in multilayer films.^{30,34} Therefore, two distinct challenges arise from the pyrolysis of mixed plastic waste. First, high oil yields have to be achieved, and second, the quality of the pyrolysis oil has to be improved for use in downstream petrochemical applications.

The most common packaging types that are sorted in most European sorting facilities are categorized according to the German Plastic Recycling or DKR system. In the Netherlands, PMD (plastic packaging, metals, beverage cartons) waste is sorted, including five plastic categories, namely, PET (bottles and trays), PE rigids, PP rigids, films/foils, and mixed plastics.^{37–40} The latter (DKR-350) includes materials left over after the more easily recyclable plastics have been removed. DKR-350 is recognized within the EU and Western Europe as a significant and challenging waste stream due to its complex composition, which hinders effective recycling through mechanical methods. Thus, DKR-350 is an ideal candidate for investigating chemical recycling. It predominantly is composed of PE, PP, PS, and PET, together with multimaterial laminates and additional components like paper, aluminum, PVDC, or EVOH.^{41,42} While a dissolution method has shown proof of concept for removing multilayer materials from DKR-350,⁴³ its economic viability remains a topic of discussion.⁴² In addition to non-PO plastics, DKR-350 often contains various contaminants, including inorganic (sand, metal) and organic (food) residues.⁴⁴ Previous work on DKR-350 pyrolysis in a fluidized bed pilot plant has reported oil yields up to 60%, which is lower than those obtained from mono streams.⁴⁵ DKR-350 in a batch fixed bed reactor showed improved liquid yields, reaching 65–69%.³⁵ The contaminations in DKR-350 were reflected in the elemental composition of the oil. High oxygen and chlorine levels were found in the plastic and propagated into the oil.³⁵ Considerable levels of other heteroatoms were also determined in the plastic, such as fluorine, silicon, and sulfur. Removing these contaminants before pyrolysis is, therefore, crucial for obtaining less contaminated products. For example, washing DKR-350 with water (and chemicals) has been shown to increase liquid yield and reduce solid and gas yields and reduce oxygen and chlorine propagation.³⁵ Boz Noyan et al. also reported that washing

Table 2. Summary of Sorting Steps and Pyrolysis Sample Preparation for the Pyrolyzed Streams in This Article in Increasing Order of Contamination from Top to Bottom

pyrolysis feedstock	sorting steps and pyrolysis sample preparation	form	
virgin plastic mix	50/50 mixing of virgin LDPE and PP	pellets	least contaminated
dry-washed DKR-350	PO-enriching of DKR-350 by NIR sorting, shredding (50 mm sieve), Loop drying (product), shredding (5 mm sieve), extruding, pelletizing (5 mm)	5 mm pellets	
unwashed DKR-350	PO-enriching of DKR-350 by NIR sorting, shredding (50 mm sieve), shredding (5 mm sieve), extruding, pelletizing (5 mm)	5 mm pellets	
loop dryer residue	PO-enriching of DKR-350 by NIR sorting, shredding (50 mm sieve), Loop drying (reject), drying (90 °C overnight)	dry 50 mm flakes	most contaminated

reduced the ash content in postconsumer plastic waste from 5.1 to 3.4%.⁴⁶

The present work aims to obtain high-yield pyrolysis liquids with low contaminant levels from real mixed plastic waste streams (DKR-350). As such, in addition to sorting, we explored (1) an economically feasible dry-washing technique to remove feed impurities and (2) the effect of process conditions, particularly the presence of added hydrogen during pyrolysis. Dry-washing, or loop drying, is a method for removing moisture and surface contaminants from plastic waste, which has been implemented in multiple companies, to improve (chemical) recycling efficiency, envisioned as more resource and energy-efficient than wet washing (e.g., a hot caustic wash) at a larger scale.^{47,48} The loop dryer works by continuously drying and centrifuging the mixed plastic waste and continuously removing surface contamination and water. Additionally, varying processing methods, such as the presence of hydrogen during pyrolysis, can further lower contaminant levels in the oil. Previous microscale studies on biomass pyrolysis have shown that hydrogen-assisted pyrolysis increases the H/C ratio of products⁴⁹ and reduces the sulfur content of S-rich coals by promoting hydrogen sulfide release.⁵⁰ To the best of our knowledge, the impact of hydrogen gas during plastic waste pyrolysis on heteroatom content has not been reported before, presenting a novel area for exploration.

2. EXPERIMENTAL SECTION

2.1. Materials. Acetone (technical, 99%) and diethyl ether (99.8%) were purchased from Riedel-de Haen. Helium 5.0, Nitrogen 5.0, Hydrogen 5.0 and calibration gas containing CO, CO₂, and C₁–C₃ were purchased from Westfalen Gassen Nederland BV. A calibration gas mixture containing 1-butene and butane was obtained from Linde Gas Benelux BV. Virgin polymers LDPE and PP were obtained from Sabic as LDPE 1922NO and PP 517P.

2.2. Methods. **2.2.1. DKR-350 Origin and Sample Preparation.** The DKR-350 mixed plastic waste was obtained from the KSI sorting facility located in Heerenveen, The Netherlands. Briefly, this facility receives mixed plastics from source-separated plastic streams, which are then sorted into different monostreams.⁵¹ Mixed plastics collected from postconsumer sources are part of a larger municipal waste management and recycling system. In general, such a waste management chain begins with household plastic waste collection through curbside recycling programs and drop-off points where mixed plastic packaging materials are aggregated. The sorting process in KSI is designed to separate materials into different streams, including PET, PE, PP, films/foils, beverage cartons, metals, and mixed plastics (DKR-350).⁵¹

In this work, about 1.4 t of postconsumer-separated DKR-350 was used. This DKR-350 was sorted, characterized for materials composition, shredded, and dry-washed using the equipment at NTCP.^{45,52} The sorting protocol using standard equipment involved removal of fines (<40 mm) in the vibrating screen, magnetic materials in the overhead magnet, and sorting using a NIR optical sorter aimed to enrich the stream for PO to produce “PO-rich DKR-350”. A two-step negative optical sorting was used to remove all other materials

but PE and PP. During sorting, the objects dropped in collection bins and small batches of samples were randomly extracted from each bin to obtain sufficient amount of representative samples for subsequent materials compositional analysis (~25 kg × 3 batches). The composition of the samples was then determined manually (see Table S1 in the Supporting Information for all categories) by weighing individual objects and determining their composition. The mass of the fractions was determined and used to calculate the composition of the feed. The PO-rich DKR-350 was shredded by using Lindner Antares 1600 industrial shredder with a 50 mm sieve to obtain the first pyrolysis stream: “unwashed DKR-350” (see Table 2). The thus obtained flakes were mixed and weighed. Part of this batch was dry-washed using a commercial Lindner Loop dryer⁵³ to reduce moisture and surface contamination to obtain a second pyrolysis stream: “dry-washed DKR-350”. In the loop drying process at NTCP, the PO-rich flakes were continuously introduced into the loop dryer (approximately 200 kg h⁻¹) equipped with rotating paddles, where flakes were dried mechanically and adhering particles were detached using controlled centrifugal force (40 Hz setting). During the process, the flakes were dried and part of the dust and small particles were mechanically separated from the main plastic material and removed through a specialized screen (7 mm sieve). The reject from the loop drying was the third pyrolysis stream and is denoted as “loop dryer residue”.

Representative samples of the PO-rich streams (2 kg, dry-washed DKR-350 and unwashed DKR-350) were shredded in a tabletop MO–DI-TECH Mini Goliath low speed granulator with a mesh size of 5 mm to grind the larger plastic flakes into a fine mixture of flakes. These flakes were thoroughly mixed and subsequently fed at a rate of 500 g/h into an MPX24 extruder with heating zones of 20 °C/175 °C/200 °C/225 °C/250 °C to be pelletized into 5 mm pellets with a Chinchio Sergio TR50 cutter. The loop dryer residue was dried overnight at 90 °C and pyrolyzed as-is. Pyrolysis of a virgin plastic mix was performed by using a 50/50 mixture of LDPE and PP. A summary of the obtained plastic feeds and their preparation is shown in Table 2.

Proximate analysis of the pellets and the flakes were carried out. The moisture content of the samples were determined by drying a 1 L container of flakes overnight in an oven at 90 °C. The flakes were sampled randomly from a bag of flakes, which was previously closed to the atmosphere. The moisture content of the plastic is heavily influenced by storage and handling prior to analysis, and further analysis of the plastic was performed on a dry basis. Pyrolysis of the plastic is therefore also performed with dry plastic pellets and flakes. For the ash content, the pellets of both PO rich DKR-350 sorted products and the loop dryer residue were calcined overnight at 500 °C. The ash was analyzed by X-ray fluorescence (XRF) by using a Malvern Panalytical Epsilon 3 X-ray fluorescence analyzer. XRF is capable of detecting elements from sodium onward; therefore, the containing solids of O, N, and F are not detected directly. Metals, silicon, sulfur, and phosphorus are reported in their oxide form, while halogens are reported as is.

2.2.2. Autoclave Pyrolysis. PO-rich DKR-350 (15 g pellets) was pyrolyzed in a Parr autoclave reactor at 420 °C for 1 h under a nitrogen or hydrogen atmosphere (1–50 bar initially). The heating rate of the autoclave was approximately 10 °C/min, and time was started upon reaching the temperature set point. After completion of the experiment, the reaction was quenched by cooling the mantle with

cold water. The product oil was filtered, and the reactor was rinsed with acetone to remove all solids from the reactor. The acetone used to rinse the reactor was filtered as well. The solid product on the filter was rinsed with acetone to remove all light products. Acetone acts as a poor solvent for waxes, causing most waxes to remain in the solid product. The solid yield was obtained by drying the filtrate overnight at 90 °C and the liquid yield was determined based on weight of the reactor before and after reaction after subtracting solid yield. The gas yield is calculated by the difference of the liquid (oil) and solid yield. In the case of pyrolysis in a hydrogen atmosphere where hydrogen is consumed during the reaction, the amount of hydrogen consumed is added to the intake material and gas, liquid, and solid yields are adjusted accordingly.

2.2.3. Pyrolysis Product Analyses. Gaseous products were analyzed by gas chromatography with a thermal conductivity detector (GC-TCD). An HP 5890 Series II gas chromatograph was used with a coupling of a Porablot Q $\text{Al}_2\text{O}_3/\text{Na}_2\text{SO}_4$ column and a molecular sieve (5 Å) column. The gas yields were determined using a calibration gas for CO, CO₂ and C₁–C₄ hydrocarbons.

The composition of the liquid product (oil) was analyzed by two-dimensional gas chromatography coupled with time-of-flight mass spectrometer (GC × GC-TOFMS). An Agilent 7890B GC with a JEOL AccuTOF GC 4G detector MS equipped with a Restek Rtx 1701 (30m × 0.25 mm × 0.25 μm) primary column and a Restek Rxi-5Sil MS (1.2m × 0.10 mm × 0.10 μm) secondary column was modulated with a Da Vinci DVLS GC2 solid-state modulator at 6 s. Samples were dissolved in diethyl ether at 2%, which acts as a better solvent than acetone used during product workup and can completely dissolve the sample. Acquisition of the MS was started after 3.5 min, coinciding with methyl hexane. The data from the GC × GC-TOFMS was further processed with a Python script to classify components according to PIONA (paraffin, iso-paraffin, olefin, naphthene, aromatic). The PIONA analysis within the visible range of the GC × GC-TOFMS was assumed to be representative for the whole oil sample and was used to calculate the yield. The method for PIONA analysis involves classifying hydrocarbons in each of the PIONA classes based on the name and the elemental formula from the library search report. The Python script is published as Script S1 in the Supporting Information.

The wax composition was determined by gas chromatography coupled to mass spectrometer (GC-MS). A sample of the solid fraction from the pyrolysis was mixed with diethyl ether to extract the soluble waxes. The sample was analyzed on an HP 6890 Series gas chromatograph equipped with a Restek Rxi-5Sil MS 30 m × 0.25 mm × 0.25 μm column and an HP 5973 Mass Selective Detector. The waxes were characterized with the NIST 05 database.

The elemental compositions of the oils were determined by elemental analysis for the elements C, H, N, S, O, and Cl. A Metrohm Model 930 Compact IC Flex Oven/SeS/PP/Deg ion chromatograph after combustion digestion on a Mitsubishi AQF-2100H was applied for Cl, and the values for CHNS were determined using an Elementar Model Vario Mikro CHNS analyzer. Oxygen content was determined by using an Elementar rapid OXY cube analyzer.

The ash and carbon content of the solid product was determined with thermogravimetric analysis (TGA) in a TGA550 apparatus from TA. A 5 mg sample was heated under inert atmosphere to 900 °C at 10 °C/min to determine the volatile solid yield (i.e., any unconverted material and heavy waxes), and the atmosphere was switched to air after 15 min at 900 °C to obtain the fixed carbon and ash yield. The elemental composition of the solid product was determined using an Elementar Vario MICRO cube CHNS. Solids were measured in duplicate and corrected to the sum of CHNS equaling 100% to correct for ash content.

3. RESULTS AND DISCUSSION

3.1. Plastic Stream Analysis. Each country or region often has its own system for categorizing plastic wastes. For example, DKR-350 mixed plastics is a useful classification within the EU especially in Germany, The Netherlands, and

neighboring countries. In this work, the exact composition of this Dutch DKR-350 material was characterized. Results show that the unwashed DKR-350 feedstock contained 51 wt % of PE and PP. The content of layered multimaterial plastics was about 14 wt %, consisting of a variety of polymers (containing about 70% PO) and others, see Table S1 (Supporting Information).⁴⁵

The composition of the DKR-350 stream after PO-enrichment by NIR sorting is given in Figure 1, and the

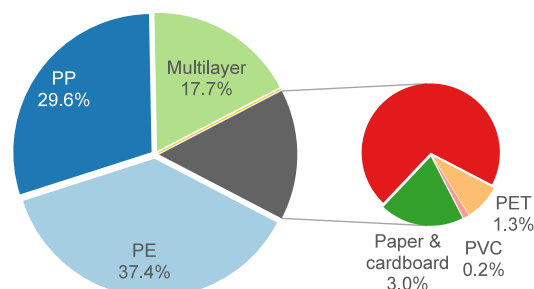


Figure 1. Object composition of PO-enriched DKR-350 determined from manual characterization.

mass balance for sorting is given in Table S2 (Supporting Information). The PE and PP content increased to 67 wt %, and the multimaterial films to 18 wt %, resulting in a total PO content of approximately 81 wt %. The PET and (E)PS content significantly decreased from 15 wt % to 1.6 wt %. These results demonstrate the effectiveness of the additional sorting step in enhancing the PO content of the DKR-350.

In the next step, the sorted PO-enriched DKR-350 was further decontaminated in a loop dryer, resulting in “dry-washed DKR-350” and a rejected “loop dryer residue” (see Table S3 in the Supporting Information). Analyses of the material composition revealed that dry-washing significantly reduced the levels of ash and water in plastic waste (Table 3). For example, the ash content decreased from 7 wt % to 4 wt % on a dry flake basis, while the loop dryer residue contained approximately 12 wt % ash. Thus, the additional dry-washing step improved the quality of the stream (in terms of PO and impurity content).

To obtain representative samples of the three DKR streams (unsorted, sorted, and dry washed; see Table 2) for laboratory-scale pyrolysis, the streams were pelletized and homogenized to minimize experimental errors due to feed inhomogeneity. Pelletizing involved shredding, extrusion, and cutting, and this increased the bulk density of the plastic to roughly 400 g/L, which is much higher than that for the bulk flake (50 g/L). The composition of the pellets was determined, revealing ash contents of 8 and 5 wt % for unwashed and dry-washed DKR-350, respectively, which is a similar reduction in ash content as compared to the flakes. In addition, shredding and pelletizing also led to a reduction in the water content, and this positively impacted the energy requirements for extruding, as the unwashed DKR-350 generated substantial amounts of steam during this step.

XRF analysis of calcined batches (500 °C, 16 h, Genlab HT75) of dry-washed DKR-350, unwashed DKR-350, and loop dryer residue was performed to determine the ash composition before and after washing. The metal oxide composition (Table 3, Tables S4 and S5 in the Supporting Information) is a function of the pretreatment method.

Table 3. Proximate Analysis of the DKR-350 Samples and the Ash Composition (XRF after Calcination at 500 °C)^b

proximate analysis	dry-washed DKR-350 (wt %)	unwashed DKR-350 (wt %)	loop dryer residue (wt %)
moisture (flakes)	1.1	7.4	12.7
ash (on pellet basis)	4.7	7.8	n/a
ash (on flake basis)	3.6	6.7	17.7
ash composition ^a			
SiO ₂	19.0	22.9	22.5
TiO ₂	29.6	19.1	4.2
CaO	19.8	19.3	21.2
Na ₂ O	8.0	14.5	20.6
Al ₂ O ₃	6.4	6.0	7.7
MgO	7.0	4.6	6.2
K ₂ O	2.2	2.7	3.4
Fe ₂ O ₃	2.4	1.8	2.7
ZnO	0.3	0.4	0.2
CuO	0.08	0.08	0.06
MnO	0.1	0.07	0.08
Cl	1.7	2.3	4.2
Br	19 ppm	73 ppm	252 ppm
P ₂ O ₅	1.6	3.2	2.7
SO ₃	1.8	3.0	4.0

^aOn an ash basis. ^bMore detailed information available in Tables S4 and S5 in the Supporting Information. Loop dryer residue ash data are the average of two analysis

Notably, the loop dryer residue had a higher Na₂O content, whereas the Na₂O levels were reduced in dry-washed DKR-350 compared to unwashed DKR. Similar but smaller changes were seen for K₂O, Cl, and Br, suggesting that the loop dryer treatment specifically affects the Na, K, Br, and Cl content. On intake, all metals in abundance above 0.1% were more prevalent in unwashed DKR-350 compared to dry-washed DKR-350, except for Ti. The discrepancy in Ti content is attributed to sampling size and variations in the large batch titanium content.

Titanium dioxide (TiO₂), used as a white pigment in plastics, varies widely in postconsumer waste and is the main source of Ti in plastics.⁵⁴ Loop drying does not remove pigments,⁵⁵ which is corroborated by the finding of Eriksen et al. that titanium is present as pigment. On the other hand, Na, K, and some of the halogens are more likely surface contaminants. This shows that the loop dryer efficiently removes certain surface contaminants from the plastic waste. Nonetheless, other impurities originating from, for example, inks and PVC will still contribute to some of the elemental impurities found in the ash of the dry-washed DKR-350.^{9,56} Dry-washing of DKR-350 shows similar improvements compared to wet-washing DKR-350 on total ash reduction, but some of the water-soluble surface contaminations are removed more readily with water washing.³⁵

3.2. Pyrolysis of Various DKR-350 Derived Feeds.

3.2.1. Product Yields. We conducted pyrolysis on the three DKR-350 derived plastic samples (dry-washed DKR-350, unwashed DKR-350, and the loop dryer residue) and a virgin plastic (50% PE and 50% PP) mixture at 420 °C in an autoclave. Liquid products were analyzed in detail. Pyrolysis of dry-washed and unwashed DKR-350 was performed in duplicate, and the average values are presented. The main product for the DKR-350 samples was a liquid (70–80%), with permanent gases (13–14%) and solids (8–15%) as byproducts

(Figure 2). The solids are defined as a combination of char, wax, and ashes. The pressure in the reactor increased from 50

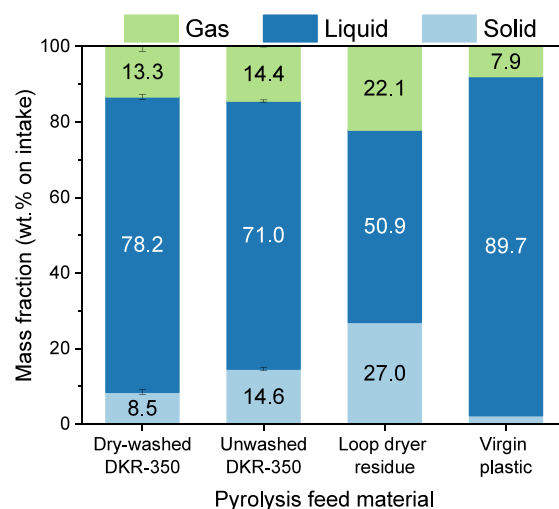


Figure 2. Yields of batch pyrolysis of DKR-350 pellets, loop dryer residue, and virgin material (420 °C, 50 bar N₂, 1 h, 15 g pellets/10 g loop dryer flakes). Results of dry-washed and unwashed DKR-350 represent the average values from experiments performed in duplicate. Gas yields by difference.

bar at room temperature to approximately 120–130 bar at 420 °C. The final pressure of pyrolysis of DKR-350 in nitrogen atmosphere after cooling was 60 bar, which is an increase of 10 bar from initial pressure. The pyrolysis oils were black, nontransparent, and slightly viscous, remaining liquid even at −15 °C. The loop dryer residue produced more gas than the DKR-350 samples and a paste of oil and solid. The virgin plastic mixture produced a yellow, transparent oil.

Significant differences in the liquid yields were observed. Dry-washed DKR-350 gave the highest liquid yield (78%), while unwashed DKR-350 gave more permanent gases and solids. Compared to a representative PO mixture of 50/50 PE/PP (90% liquid yield), sorted DKR-350 (unwashed and dry-washed) produced fewer liquids and more solids and gases. This difference is attributed to the presence of non-PO plastics (see Table S1 in the Supporting Information), additives, contaminants, and high amounts of ash in the DKR-350 feeds. Previous work by Genuino et al. reported a 13.7% gas, 65.9% liquid, and 13.2% solid yield from the pyrolysis of unsorted DKR-350 in a batch, fixed bed reactor setup, indicating a nearly 19% enhancement in oil yield for dry-washed, sorted DKR-350 in this work.³⁵

Compared with pyrolysis data reported from other mixed plastic waste sources, the oil yield in this work is relatively high. Kaminsky et al. obtained about 50% oil from German household mixed plastic waste with a 75% polyolefin content and high PS content in a fluidized bed reactor.⁵⁷ More recently, Singh et al. obtained a maximum of 84% oil from PVC-free, manually sorted and cleaned plastic waste.¹⁶ Previous work with DKR-350 also yielded 66% oil, which increased to 77% with extensive cleaning, which is exceeded here with a much simpler cleaning process.³⁵ Generally, clean mixed plastics yield around 85% oil, but impurity levels and wax formation vary significantly.^{25,58} Achieving as high as 78% oil yield from heavily contaminated plastic waste without

manual cleaning indicates the effectiveness of the mechanical pretreatment steps.

A more subtle variation is observed in the product composition when comparing unwashed and dry-washed DKR-350. Dry-washed DKR-350 consistently gave 42% reduction (−6%pt) solid yield and slightly less gas, resulting in a more than 10% increase (+7%pt) in liquid product from unwashed DKR-350. Additionally, the product oil was easier to separate from the solids due to less filter clogging.

The pyrolysis of a 50% PE/50% PP virgin plastic mixture produced 90% oil. Unlike DKR-350 pyrolysis, the solid from the virgin plastic mixture was in the form of wax, which was also seen in previous observations.⁵⁹ The combined wax and oil yield therefore exceeds the reported 90% liquid oil yield.

The loop dryer residue, containing less PO than the other DKR-350 streams, produced 51% oil with high amounts of CO, CO₂, char, and ash. The loop dryer residue, estimated to contain approximately 50–60% polyolefins compared to approximately 75–80% for DKR-350 after correcting for surface contamination, resulted in a paste-like liquid product that was harder to separate due to the higher ash content (Table 3). Due to the high oxygen content in the loop dryer residue, it may be more suited for gasification than pyrolysis.

The gas compositions of the pyrolysis products for the dry-washed DKR-350, unwashed DKR-350, and loop dryer residue differed considerably (see Figure 3). Unwashed and dry-

allows for the separation of volatile components over a range of boiling points and polarities. For example, effectively separation of aromatics and nonaromatics, even when boiling points are very similar, is possible (Figure S2 in the Supporting Information). The GC × GC-TOFMS results also identified contaminants such as alcohols, aryl and alkyl nitriles, caprolactam, ketones, and phenol and phenol derivatives such as cresol in the pyrolysis oil from the different DKR-350 streams. These are most likely derived from non-PO plastics or other contaminants.

GC × GC-TOFMS has previously been applied for PIONA analysis in kerosene by analyzing specific ionic fragments.⁶¹ Using a Python algorithm with library search data, we successfully applied a PIONA classification to the oil, and the results are given in Figure 4. The major components in the oils are linear paraffins, followed by naphthenes and olefins. Aromatic compounds are present in low amounts in all pyrolyzed samples (<12% in dry-washed, unwashed DKR-350, and 50% PE/50% PP virgin plastic). Sensitivity analysis of the PIONA analysis concentration and integration methods shows consistent results (Figure S3 in the Supporting Information).

In the oil from dry-washed DKR-350, paraffins are the major constituent (32%). Olefins make up 19% of the oil and is found in similar quantities as naphthenes. The amount of olefins is lower for unwashed DKR-350 (17%). The aromatic content also decreased from 12% to 9%. The unwashed DKR-350 has a different iso-paraffin to paraffin ratio compared to dry-washed, likely due to subtle compositional differences in the plastic. Similar levels of aromatics (~9%) but higher amounts of olefins (>50%) and lower amounts of naphthenes (<10%) were reported by Dukle et al. for plastic pyrolysis oil from undisclosed sources.⁶² Saturated naphthenes and olefins have identical C/H ratios, which suggests potential difficulties in distinguishing olefins from cyclic alkanes using GC × GC-TOFMS.

For the pyrolysis product of a 50/50 mixture of virgin PP and PE, 35.7% of the GC × GC chromatogram area consisted of paraffins, 16.6% of iso-paraffins, 17.3% of olefins, 29.3% of naphthenes, and 1.1% of aromatics. This composition significantly differed from unwashed and dry-washed DKR-350, which contained notably more aromatics. This suggests that the aromatics likely originated from other plastics in the feed (e.g., the formation of benzene, ethylbenzene and styrene from PS) or from catalytic effects of ash in plastic waste and the 1.6% of aromatic plastics (PS/PET) and 17.7% multi-material films in DKR-350 likely contributed to the higher aromatics content.⁴¹ Unclassified materials in DKR-350 (7%) also potentially contained aromatics. The higher linear paraffin content in DKR-350 compared to the PE/PP mix is most likely due to a polyethylene to polypropylene ratio, though the exact PE/PP ratio in DKR-350 is unknown due to the multimaterial variable content.

3.2.3. Elemental Compositions of Dry-Washed and Unwashed DKR-350 Pyrolysis Products. The elemental composition of the oils from dry-washed and unwashed DKR-350 was determined for C, H, N, S, O, and Cl (Table 4), which are major contaminants of interest for many downstream processing.^{63,64} The nitrogen is most likely from polyamides or polyurethanes present in the mixed polyolefin stream.¹⁵ Alternatively, it may also originate from degradation products of nitrocellulose in printing ink, as nitrocellulose already decomposes during extrusion.⁶⁵

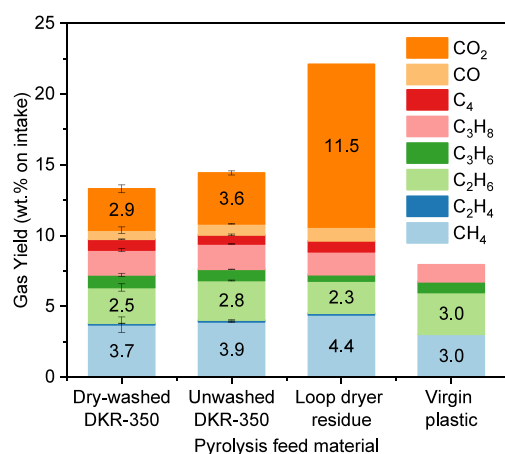


Figure 3. Gas yields and composition from the pyrolysis of dry-washed DKR-350, unwashed DKR-350, loop dryer residue, and a 50% PE/50% PP virgin plastic mixture (batch pyrolysis, 420 °C, 50 bar N₂, 1 h, 15 g pellets/10 g loop dryer flakes). Gas composition is determined by GC-TCD. Results of dry-washed and unwashed DKR-350 represent the average from experiments performed in duplicate. Total gas yields by difference.

washed DKR-350 gave similar gas yield and composition (13–14%), primarily methane, CO₂, ethane, and propane. On the other hand, the loop dryer residue produced by far more gas phase products (22%), with a notable increase in CO₂ (11.5% on intake). This result suggests that the loop dryer removes oxygen-rich material such as PET or surface contaminants such as biomass and food residue, which are known to produce CO₂ upon pyrolysis.⁶⁰

3.2.2. Liquid Phase Analyses. Two dimensional gas chromatography coupled with a time-of-flight mass spectrometer (GC × GC-TOFMS) is very efficient for detailed molecular analyses of complex pyrolysis derived liquids. It

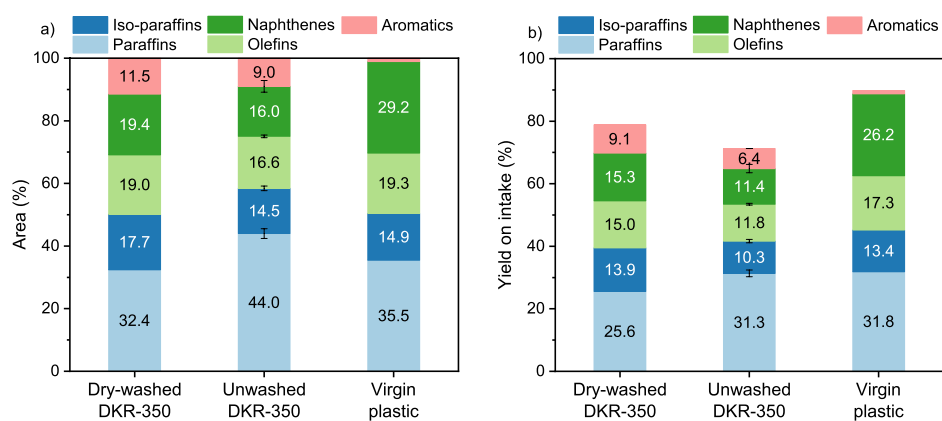


Figure 4. Oil composition of the pyrolysis liquids from dry-washed DKR-350, unwashed DKR-350 and 50/50 LDPE/PP classified into paraffin, iso-paraffin, olefin, naphthene, and aromatics (PIONA) by GC $\times$ GC-TOFMS data processing, based on oil (a) and on intake (b), by multiplying oil yield with composition. Analysis of unwashed DKR-350 was based on duplicate measurements.

Table 4. Elemental Composition (C, H, N, S, O, Cl) and Yield of Dry-Washed and Unwashed DKR-350 Oil in Batch Pyrolysis (420 $^{\circ}$ C, 50 bar of N₂, 1 h)

	C (%)	H (%)	N (%)	S (%)	O (%)	Cl (ppm)	others (%) ^a	atomic O/C ^a	atomic H/C ^a
oil composition									
dry-washed DKR-350	77.12	12.52	0.99	0.12	6.86	32	2.39	0.07	1.93
unwashed DKR-350	78.66	13.08	0.65	0.03	6.56	17	1.03	0.06	1.98
oil yield									
dry-washed DKR-350	60.31	9.79	0.77	0.09	5.37	25	1.87		
unwashed DKR-350	55.85	9.28	0.45	0.02	4.66	12	0.73		

^aOther obtained by difference from all identified elements. O/C and H/C ratio calculated from C, H and O content. More detailed information including error margins in Table S6.

The oils had a combined C and H content of 90–92%, with the remainder mainly oxygen and nitrogen, and smaller amounts of chlorine and sulfur. “Others” comprised 1–3% of the content, which was the unidentified part of the oil. The H/C ratio was slightly higher for unwashed DKR-350, likely due to increased aromatic components in dry-washed DKR-350, as also shown by GC $\times$ GC-TOF-MS analyses (*vide supra*).

The oxygen content in the oil from unwashed DKR-350 (6.56%) was slightly lower than that from dry-washed DKR-350 (6.86%). The O/C ratio was comparable at 0.06 and 0.07, respectively. Compared to clean plastic mixes, the oxygen content of 6–7% is considered too high for petrochemical applications and would have to be lowered with more rigorous pretreatment.¹⁵ In contrast to previous experiments with DKR-350, which had 14.5% and 8.2% for unwashed and wet-washed DKR-350, the oxygen is lower.³⁵ The difference is mainly attributed to the sorting to PO-rich fraction, which reduced the PET and paper/cardboard content and therefore the oxygen content of the DKR-350.

Chlorine content in the pyrolysis oil ranged from 17 and 32 ppm in unwashed DKR-350 and dry-washed DKR-350, respectively. Complete chlorine removal from the feed material by washing is nearly impossible as chlorine is present in many plastic wastes, such as multimaterial films containing PV(D)C or as a pigment.⁹ It is suspected that the presence of char and ash may facilitate chlorine absorption.^{66,67} The chlorine content (17–32 ppm) in pyrolysis oil of sorted DKR-350 from this work is 5–16 times lower than reported by Genuino for the oil obtained by pyrolysis of mixed waste plastics using a fixed bed reactor (275 ppm for DKR-350 and 151 ppm for wet-washed DKR-350).³⁵ It is also suggested that long reaction

times aid in removing chlorine-containing components from the oil.

Nitrogen content in the oils from dry-washed DKR-350 and unwashed DKR-350 was below 1%, with dry-washed DKR-350 having the highest nitrogen content (0.99%) and unwashed DKR-350 the lowest (0.65%). Similar nitrogen contents have been found in pyrolysis oil of mixed polyolefin contaminated with other polymers, traced back to polyamide or polyurethane contamination.¹⁵ A small amount of sulfur was also detected in the pyrolysis oil, with dry-washed DKR-350 having the highest sulfur content (0.12%) and unwashed DKR-350 the lowest (0.03%). Kusenberget al. found no detectable sulfur content and an oxygen content of 0.32 wt % in a pyrolysis oil from mixed plastics (mixed PO with contaminations of other plastics).¹⁵ This is not surprising as their feed was subjected to severe pretreatment steps including washing, float-sink separation, and extrusion, resulting in considerably cleaner feed than the DKR-350 used in this work.

The pyrolysis oil obtained from both DKR-350 feedstocks is too contaminated for direct use in petrochemical industries.³³ Post-treatment, such as hydrotreatment, to reduce the heteroatom and unsaturated content, is required to get the pyrolysis oil closer to naphtha specifications. Without any post-treatment, it may only have limited applications for blending with fossil streams.³⁴

The solid residues were also analyzed for elemental composition (Table S7 in the Supporting Information). The solid residue from the unwashed DKR-350 was richer in hydrogen, nitrogen, and sulfur than the char from dry-washed DKR-350, suggesting that the solid residue acts as an absorbent for impurities.^{68,69} The increased absorption of

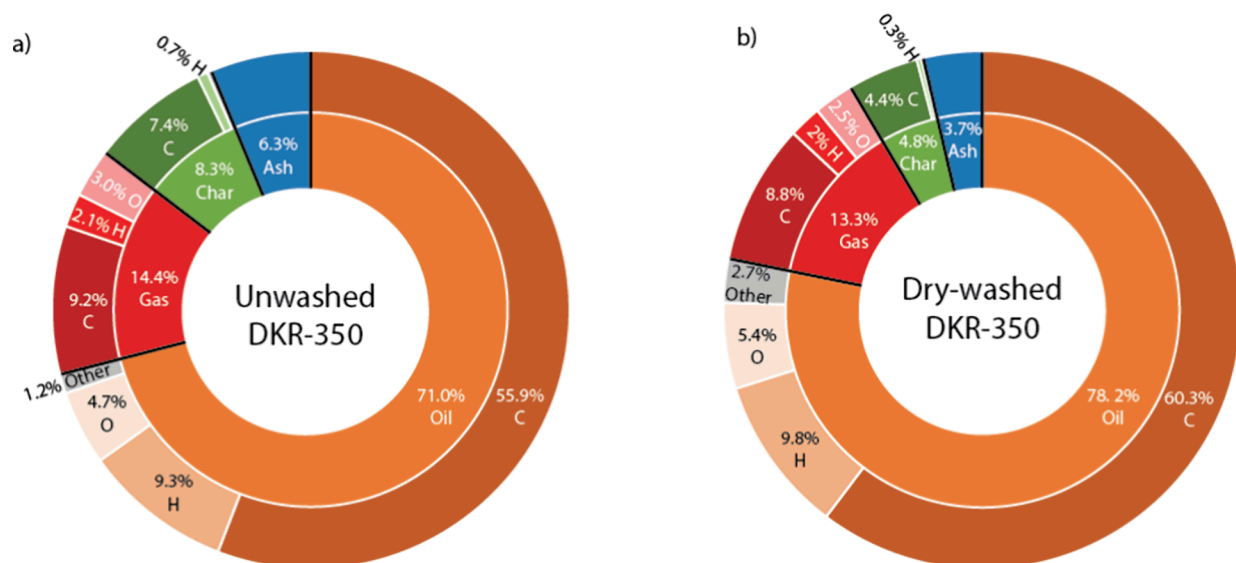


Figure 5. Mass balance (inner rings) and C, H, O elemental composition per phase and overall elemental composition of the intake material of the pyrolysis products (outer rings) of unwashed DKR-350 (a) and dry-washed DKR-350 (b) in batch pyrolysis (420 °C, 50 bar N₂, 1 h, 15 g pellets). N, S, Cl, and other components classified as Other. Liquid yields determined as average over multiple runs.

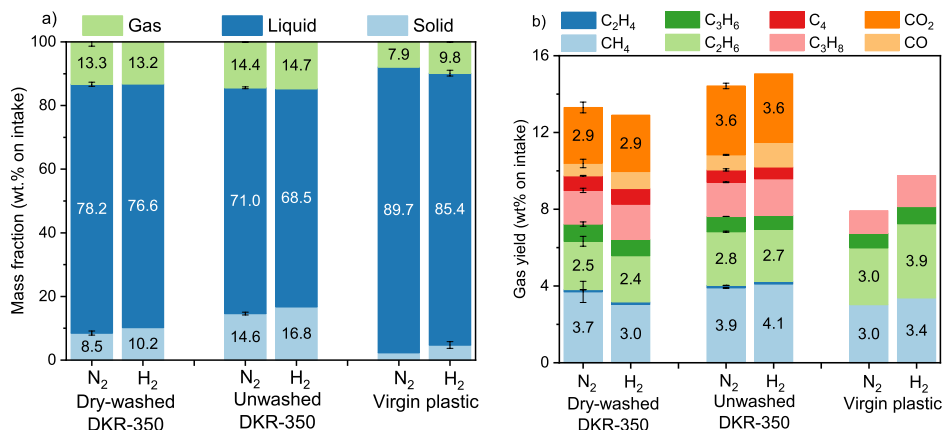


Figure 6. Mass balance (a) and gas composition (b) for unwashed and dry-washed DKR-350, and virgin material mix in presence of hydrogen and nitrogen. Mass balance for pyrolysis in hydrogen calculated on the basis of consumed hydrogen and total plastic intake. Gas yields by difference.

heteroatoms in the solid residue of unwashed DKR-350 is in line with the finding of reduced heteroatom content in the oil. Thus, we conclude that dry-washing significantly increased the carbon and hydrogen yield toward the oil upon pyrolysis. Nevertheless, oxygen, nitrogen, and sulfur contents are higher than unwashed DKR-350 due to less absorbance in the solid residue.

3.2.4. Elemental Balance of Dry-Washed and Unwashed DKR-350. On a total elemental composition basis, the pyrolysis products of the dry-washed and unwashed DKR-350 were similar. The higher sulfur and nitrogen content in dry-washed pyrolysis oil were offset by more absorption of these components in the char from unwashed DKR-350 pyrolysis products. Figure 5 shows the elemental composition of each phase and the elemental composition of the product total. The total carbon, hydrogen, and oxygen content of the different DKR-350 samples was similar, albeit slightly higher in dry-washed DKR-350 due to the reduced ash content. This indicates that the main effect of loop drying is the removal of surface contamination, such as sand and biomass, with the same plastic composition (consistent with characterization

data on feed and pyrolysis products). The increased CO and CO₂ from unwashed DKR-350 correlated well with biomass/biogenic surface contamination.

Thus, we conclude that dry-washing of sorted DKR-350 has a very positive effect on the liquid yield. The liquid yield is higher than for unwashed DKR-350, with hydrocarbon amount in the oil increasing from 65.1% to 70.1%. As a consequence, the carbon efficiency (defined as the ratio of carbon in oil over total carbon in gas, solid, and oil) increases from 77.0% to 82.0%. However, the oil obtained from dry-washed sorted DKR-350 contains slightly more heteroatoms, such as O, N, and S.

After determining the elemental compositions for all phases from unwashed and dry-washed DKR-350 pyrolysis, the elemental composition of the intake material was determined. Unwashed and dry-washed PO-rich DKR-350 have similar object composition, and the ash-free elemental composition is also similar. Using the elemental balances from pyrolysis of dry-washed and unwashed PO-rich DKR-350 in nitrogen and atmosphere, an elemental composition of PO-rich DKR-350 is obtained. On an ash-free basis, PO-rich DKR-350 consists of

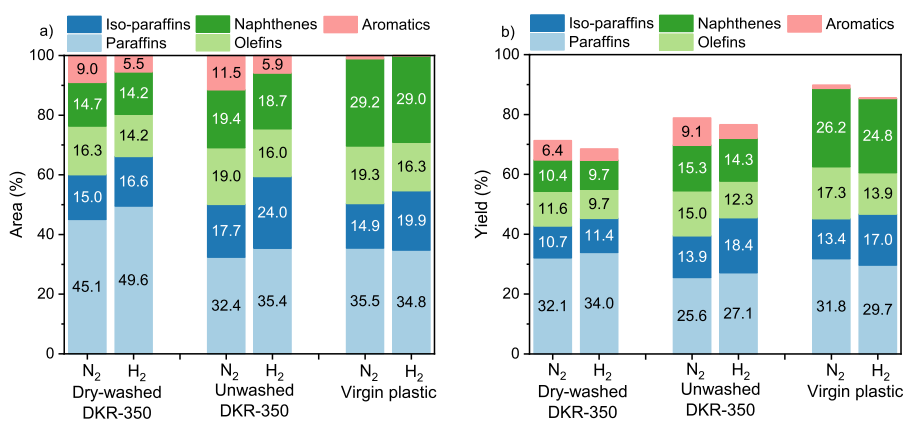


Figure 7. Composition of the liquid phase after pyrolysis of dry-washed DKR-350, unwashed DKR-350, and 50/50 PE/PP feed pyrolysis oil (420 °C, 50 bar H₂ or N₂, 1 h) classified into paraffin, iso-paraffin, olefin, naphthene, and aromatics (PIONA) by GC × GC-TOFMS data processing, based on oil (a) and on intake (b), by multiplying oil yield with composition.

76.4–79.8% C, 12.4–12.9% H, 6.7–8.2% O, and 0.4–2.7% others.

The other fraction is a combination of nitrogen, sulfur, and unidentified components. Nitrogen is present in the oil and solid in quantities of approximately 1% of the intake material. However, note that nitrogen-containing compounds in the gas product were not measured.

Heteroatoms from the pyrolysis feedstock will propagate into the pyrolysis oil product. Removing heteroatoms from the plastic waste will reduce the heteroatoms content of the oil.³⁵ Dissolution of multilayers, thorough sorting, or thorough washing will enhance heteroatom removal but will come at the cost of reducing the amount of plastic suitable for chemical recycling by pyrolysis.^{17,34} If chemical recycling is to be applied to the majority of plastic waste, it is necessary to apply post-treatment before being fit for petrochemical applications. Similarly, dry-washing here is established as an application of plastic waste to enhance the carbon yield of chemical recycling at the cost of reducing the total volume of suitable waste.

3.2.5. The Effect of Hydrogen-Assisted Pyrolysis on the Product Distribution. Another approach to reduce the amount of impurities in the product oil was also investigated and involves the presence of hydrogen during pyrolysis. Pyrolysis experiments were performed in the presence of hydrogen (50 bar) under otherwise similar conditions to those for experiments with nitrogen (420 °C, 1 h, 15 g unwashed DKR-350 and dry-washed DKR-350). Initial experiments at 50 bar hydrogen pressure showed a slight reduction in liquid yield for both dry-washed and unwashed DKR-350 compared to a nitrogen atmosphere (71.0% under N₂, 68.5% under H₂ for unwashed), with no visual difference in oils (Figure 6). However, solids yield increased to 10.2% and 16.8% for unwashed DKR-350 and dry-washed DKR-350 whereas the gas yield remained similar. The increase in solid yield suggests more char or wax formation, as the inorganic part of the solid residue (ash) was constant. For pyrolysis experiments under nitrogen, the pressure at the end of the batch reaction (at 420 °C) was about 10 bar higher than the initial pressure at 420 °C due to the formation of permanent gases (Figure 6). This was not observed for hydrogen, where the pressure measurements at the start and end were about similar. This implies that hydrogen is consumed during pyrolysis, also evidenced by considering the liquid and solid phase composition (*vide infra*). Similar trends in product yields were also observed for

the 50% PP/50% PE mixture when hydrogen was added to the pyrolysis reaction. Here, a slightly lower liquid yield was found in combination with more solid formation (Figure 6).

The solid composition for virgin plastic pyrolysis in the presence of hydrogen is a wax, which is similar to the solid composition of virgin plastics in pyrolysis in a nitrogen atmosphere. Pyrolysis in a nitrogen atmosphere yields approximately 2.3% wax, while pyrolysis in a hydrogen atmosphere yields approximately double (4.8%). As olefins with similar carbon numbers have a lower melting point than their paraffin counterpart, hydrogenation of olefins during pyrolysis in hydrogen atmosphere causes the formation of more solid product.

GC × GC-TOFMS analysis revealed distinct differences in oil composition between pyrolysis in a hydrogen and nitrogen atmosphere. In the pyrolysis with a hydrogen atmosphere, a lower aromatic content in the oil was found, as shown in Figure 7. Two pathways can be envisioned for aromatics formation, viz., (i) from pyrolysis of aromatic containing plastics in the feed such as PET or PS and (ii) formation of aromatics during the pyrolysis process. As the feed compositions are similar for the nitrogen and comparative hydrogen experiments, the observation that less aromatics are formed in the presence of hydrogen is due to reactive intermediate stabilization during pyrolysis. This is observed for both the oil composition and the aromatic yield based on material intake. The literature for pyrolysis studies using coal in a downer pyrolyzer and a batch reactor has shown similar results, and it is assumed that hydrogen stabilizes reactive intermediates, which in turn leads to the formation of more liquid product.^{70,71}

Furthermore, the unsaturated content in the form of olefins was reduced by hydrogen-assisted pyrolysis. The olefin content is reduced in both unwashed DKR-350 (16.3% to 14.2%) and dry-washed DKR-350 (19.0% to 16.0%). The saturated iso-paraffins and paraffins were found to be present in a higher amount for both unwashed DKR-350 and washed DKR-350, showing that the hydrogen is involved in hydrogenation of olefins to paraffins.

In the virgin PE/PP mix, the unsaturated content of the oil prepared under a hydrogen atmosphere (16.5%) is slightly lower than in the presence of nitrogen (20.5%). These findings are in line with that of the unwashed and dry-washed DKR-350, again confirming that the presence of hydrogen reduces

Table 5. Elemental Composition (C, H, N, S, O, Cl) of Dry-Washed and Unwashed DKR-350 Oil at Varying Pressures and Atmospheres (420 °C, Flushed with or Pressurized with 50 bar H₂ or N₂, 1 h)

	C (%)	H (%)	N (%)	S (%)	O (%)	Cl (ppm)	other (%) ^a	atomic O/C ^a	atomic H/C ^a
unwashed DKR-350, 50 bar N ₂ ^b	78.66	13.08	0.65	0.03	6.56	17	1.03	0.06	1.98
unwashed DKR-350, 50 bar H ₂	81.48	13.13	0.47	0.09	4.42	20	0.43	0.04	1.92
dry-washed DKR-350, 50 bar N ₂ ^b	77.12	12.52	0.99	0.12	6.86	32	2.39	0.07	1.93
dry-washed DKR-350, 50 bar H ₂	79.01	12.9	0.57	0.11	5.69	26	1.69	0.05	1.95

^aOther obtained by difference from all identified elements. O/C and H/C ratios were calculated from C, H and the O content. More detailed information including error margins in Table S8. ^bShown previously in Table 4.

Table 6. Elemental Composition of the Solid Residue from Pyrolysis of Unwashed DKR-350 in Batch Pyrolysis (420 °C, 50 bar H₂ or N₂, 1 h), on an Identified Basis and on a Char Basis (C + H + N + S = 100%), to Account for Ash content^a

	C (%)	H (%)	N (%)	S (%)	atomic N/C	atomic S/C (× 10 ⁻³)	atomic H/C
solid composition							
unwashed DKR-350, H ₂	87.52	10.9	0.94	0.64	0.01	2.7	1.48
unwashed DKR-350, N ₂	89.25	7.83	1.97	0.95	0.02	4.0	1.05
solid yield							
unwashed DKR-350, H ₂	9.59	1.19	0.10	0.07			
unwashed DKR-350, N ₂	7.41	0.65	0.16	0.08			

^aYields and composition based on duplicate experiments. More detailed information, including error margins, is in Table S9.

the unsaturated content. This also coincides with the finding of increased wax yield, as more saturated (i.e., heavier) compounds are detected.

The effect of the hydrogen atmosphere was further explored with pyrolysis experiments conducted at lower pressure 1 bar (instead of 50 bar used in the experiments above for both nitrogen and hydrogen) for unwashed DKR-350. For these experiments, the autoclave reactor was not pressurized with 50 bar of nitrogen or hydrogen but instead flushed with nitrogen or hydrogen and starting the reaction as atmospheric pressure. The product distribution and the PIONA results are shown in Figures S5 and S6 in the Supporting Information. These experiments show that even a small amount of hydrogen (1 bar under starting conditions) has a significant influence on the liquid phase composition even at low pressure. Hydrogen is suggested to be involved in reduction of unsaturated content via hydrogenation of, for example, olefins. Here, catalytically active metals in the feed such as nickel, iron, and calcium, may play a role.⁷²

3.2.6. Elemental Compositions of Dry-Washed DKR-350 and Unwashed DKR-350 in a Hydrogen Atmosphere. Elemental analysis of oils obtained at different hydrogen and nitrogen pressures quantified the effect of hydrogen during unwashed DKR-350 pyrolysis. The atomic oxygen to carbon ratio is a parameter that was determined from the elemental analysis to describe the removal of oxygen containing impurities by hydrogen assisted pyrolysis. The oxygen content of the oil from unwashed DKR-350 was reduced from 6.56% under 50 bar nitrogen to 4.42% in 50 bar hydrogen, corresponding to an O/C ratio decrease from 0.06 to 0.04 (Table 5). A similar trend was seen for dry-washed DKR-350, with the O/C ratio reduced from 0.07 to 0.05, suggesting hydrogen reduces heteroatoms in the oil.

Nitrogen content followed a similar trend, with the highest nitrogen contents observed in a nitrogen atmosphere (up to 1% in dry-washed DKR-350) and the lowest in hydrogen-assisted pyrolysis (down to 0.5% for 50 bar H₂ with unwashed DKR-350).

The PIONA analysis previously showed a reduction in the unsaturated content with hydrogen-assisted pyrolysis. The H/

C content was obtained from elemental analysis for dry-washed DKR-350 in hydrogen-assisted pyrolysis and pyrolysis in a nitrogen atmosphere. This shows an increase that is in line with an increased saturation content as the H/C ratio increased from 1.93 to 1.95.

However, the H/C ratio for unwashed DKR-350 decreased (1.98 to 1.92) in hydrogen-assisted pyrolysis; therefore, that would suggest different phenomena playing a role. A part of the oil for unwashed and dry-washed DKR-350 is not quantified ("Other" in Table 5) and ranges from 0.4% to 2.4%. The trend of the H/C ratio varying with atmosphere type is dependent on small differences in hydrogen and carbon contents in the oil. With a part of the oil being unidentified, the error margin in the elemental analysis causes the observed trend to not be significant enough to fully establish the hydrogenating effect of hydrogen by itself. However, in combination with other analyses, the hydrogenating effect is shown. In addition, the level of impurities still exceeds typical steam cracker thresholds³³.

Pyrolysis of DKR-350 in a hydrogen atmosphere produced more solid yield than pyrolysis in a nitrogen atmosphere. TGA of the solid residues was used to investigate what types of components were formed, such as ash, fixed carbon, or volatile solid products. The thermogram showed that more volatile solid components (i.e., heavy waxes) are present in the solid residue from unwashed and dry-washed DKR-350 in a hydrogen atmosphere compared to the nitrogen atmosphere. The most noticeable difference between the thermograms of dry-washed and unwashed DKR-350 solid residue is between 300 and 450 °C. In that range, the solid obtained by pyrolysis in a nitrogen atmosphere loses 5% of its initial mass, while the solid obtained in an hydrogen atmosphere loses approximately 30%. Hydrogen-assisted pyrolysis therefore produces more volatile solid products. This boiling point range corresponds roughly to paraffins in the range of C₁₇–C₃₀, which are solid at room temperature. The diethyl ether extracts of the solid residue from washed and unwashed DKR-350 pyrolyzed in a hydrogen and nitrogen atmosphere were analyzed with GC-MS. This showed paraffin wax being present in the solid residue. The extract in nitrogen pyrolyzed DKR-350 solids was

not saturated with waxes, whereas the hydropyrolyzed DKR-350 still showed undissolved waxes in solution. The wax content was therefore higher in the case of pyrolysis in a hydrogen atmosphere.

The CHNS elemental composition of solid residues was also determined to further investigate the effect of hydrogen on the pyrolysis products (Table 6). Results show that the solid residue from pyrolysis under hydrogen is richer in hydrogen and carbon than the solids from pyrolysis of DKR-350 under a nitrogen atmosphere, in addition to yielding slightly more solid. The atomic H/C ratio is also higher in the case of hydrogen assisted pyrolysis, which corresponds to an increase in the wax content in the solid residue.

Thus, the observation that significant amounts of hydrogen are consumed for experiments under a hydrogen atmosphere is due to a combination of factors. From both elemental analysis (solid and liquid) and the PIONA analyses (liquid), it becomes clear that hydrogen is incorporated into the solid product and hydrogenates unsaturated and oxygenated species in the oil. The addition of hydrogen leads to the formation of slightly more wax, which is most likely caused by the stabilization of reactive intermediates. A reduction in heteroatoms (N, S, O, Cl) was observed in char and oil, most clearly for N and O. As these components in the oil are present in low quantities regardless of the pyrolysis atmosphere and with small variations between experiments, a clear effect was not established for S and Cl.

4. CONCLUSIONS

In this study, we investigated the pyrolysis of (pre)treated postconsumer-separated mixed plastic waste streams, providing critical insights into the effects of preprocessing on the yields and quality of the resulting pyrolysis oil. We demonstrated that an additional PO-enriching sorting step (performed at an industrial scale) significantly increases oil yields due to the higher polyolefin content in plastic waste. Furthermore, dry-washing, which has emerged as a highly efficient technology for removing surface contaminants such as ash and halogens, led to increased liquid yields and reduced solid and gas yields upon pyrolysis. Additionally, dry-washing removes water from the feedstock, making it more suitable for pelletizing, a preferred pretreatment step for larger-scale operations as it increases the feedstock density by an order of magnitude. The dry-washed material yielded ~10% more liquid pyrolysis products than the unwashed material and exhibited improved carbon efficiencies. However, the increased ash and char production from unwashed DKR-350 absorbed impurities (N, S, O, and Cl) during pyrolysis, resulting in lower impurity levels in the unwashed DKR-350 pyrolysis oil. By demonstrating the effects of removing contaminants, enhancing PO content, or introducing additional moisture removal and decontamination steps, such as dry-washing, this work provides insights that may be relevant beyond the waste classification context. For example, this approach could inform best practices for pretreatment of specific waste streams for different methods of chemical recycling, including materials composition characterization, with the aim of optimizing oil yields and composition.

Processing under a hydrogen atmosphere increased the saturated hydrocarbon content of the liquid, enhancing the quality of the oil as a potential refinery feed. Hydrogen presence reduced unsaturated hydrocarbons and particularly aromatics in the oil, as observed in the GC × GC-TOFMS

analysis. Although the exact mechanism of hydrogen addition during plastic pyrolysis is not fully understood, it was found to produce oils with significantly lower heteroatom content (oxygen, nitrogen, and sulfur). Sulfur and chlorine levels of the pyrolysis oil were low across all methods, promising for downstream applications. The lowest impurity levels were obtained from the pyrolysis of unwashed DKR-350 in the presence of hydrogen (4.4% O, 0.5% N, 20 ppm Cl, and 0.09% S). Despite this, the high oxygen, nitrogen, chlorine, and sulfur still warrants hydrotreating before further downstream processing in the petrochemical industry.

This work shows that dry-washing is a method to improve the efficiency of pyrolysis, similar to thorough sorting or aqueous washing. As is the case with these quality improvement methods, this comes at the cost of losses in the waste volume. Further research is needed to assess the impact of scaling up and conduct a life cycle assessment of the dry-washing method for producing pyrolysis oil from mixed plastic waste. This includes evaluating the benefits of increased oil quantity while reducing the feedstock volume. Finally, the influence of hydrogen-assisted pyrolysis and its exact mechanisms could be further explored by using model studies with metal oxides to simulate ash and virgin polymers.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.energyfuels.4c04745>.

DKR-350 material composition prior to and after sorting to PO-rich DKR-350, mass balances of the sorting and dry-washing, detailed ash composition and metal/halogen content of the plastic waste, GC × GC-TOFMS chromatogram of a typical oil, PIONA sensitivity analysis, detailed elemental composition and yields of pyrolysis oils and pyrolysis solid residue for dry-washed and unwashed DKR-350, detailed elemental composition and yields of pyrolysis oils and pyrolysis solid residue for varying gas types and pressures, ash free mass and elemental balance of unwashed and dry-washed DKR-350, mass balances and PIONA analysis for pyrolysis products of varied pressures and gas types, graphic H/C and O/C content over varying atmospheric conditions, thermograms of DKR-350 solid residue from dry-washed and unwashed DKR-350 pyrolyzed in varying atmosphere, complete mass balance and gas yields of pyrolysis of dry-washed DKR-350, unwashed DKR-350, loop dryer residue and virgin plastics, and the PIONA analysis script (PDF)

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